

Introduction to Statistical Methods

(S2-25 AIMLCLZC418) – Assignment 1

AIML – Section 7

Question 1&2 carries 2.5 marks each.

Question 3 carries 5 marks

Total - 10 Marks

Duration: 30th May 2026 – 14th June 2026

1) Submissions are individual

2) Solve these on paper, scan, and upload

3) Plagiarism results in zero marks

4) Write your name, BITS ID and Section on each page

5) Only handwritten solutions with formula, full steps with proper justification are required.

Answer all the questions

1. A manufacturing unit recorded the number of defective items produced per month over a year as follows:
12, 15, 14, 13, 16, 18, 14, 15, 13, 14, 17, 14, 36
 - (a) Compute, mean, median and mode of the defects. Comment on whether the distribution is symmetric or skewed.
 - (b) Give the five – point summary and find the outliers if any using IQR method.
 - (c) The company wants to report monthly performance to clients. Should they report mean or median? Justify mathematically. **[2.5 M]**
2. A company wants to predict whether a website visitor will subscribe to their newsletter based on one feature: Device type. You are given the following training data:

Visitor	Device	Subscribe
1	Mobile	No
2	Mobile	Yes
3	Desktop	Yes
4	Desktop	No
5	Tablet	Yes

- (i) Compute the prior probabilities: $P(\text{Subscribe} = \text{Yes})$, $P(\text{Subscribe} = \text{No})$
- (ii) A new visitor using a Mobile device. Using the Naïve Bayes Classifier, predict whether they will subscribe. **[2.5 M]**

3. A university offers three certificate courses to first-year students: Data Science (D), Cyber Security (C), and Business Analytics (B). In a survey of 500 students, 230 students enrolled in Data Science, 180 enrolled in Cyber Security, 150 enrolled in Business Analytics, 90 enrolled in both Data Science and Cyber Security, 70 enrolled in both Data Science and Business Analytics, 60 enrolled in both Cyber Security and Business Analytics, and 35 enrolled in all the three courses.

Represent the given data pictorially and, if one student is selected at random, answer the following:

1. Find the probability that the student enrolled in at least one of the three courses.
2. Find the probability that the student enrolled in exactly two courses.
3. Find the conditional probability that a student enrolled in Cyber Security given that the student enrolled in Data Science.
4. Are the events D and C independent, justify mathematically.
5. Interpret the result obtained in part (3) in the context of students' course preferences.

[5 M]

----- ALL THE BEST -----